Yangruirui ZHOU

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EDUCATION

Boston University, Boston, MA 09/2020-01/2026

Doctor of Philosophy (Ph.D.) in Computer Engineering

GPA: 3.91/4.00

Advisor: Douglas Densmore

Thesis: AI-driven design automation for microfluidics and multicellular systems

Research Focus: AI-enabled microfluidic CAD, formal verification, and multicellular biological system design.

University of Electronic Science and Technology of China (UESTC), Chengdu, China 09/2016-06/2020

Bachelor of Engineering in Software Engineering *(Elite program)*

GPA: 3.94/4.00 (Rank: 6/740)

Thesis: Analysis of Postoperative Applications of Pose Tracking Algorithms.

AWARDS & HONORS

iGEM "**Gold medal**" (Student Mentor, wiki: https://2024.igem.wiki/bostonu/) 11/2024

China Youth Science and Technology Innovation Award (Nominated by UESTC) 05/2020

Most Outstanding Students of UESTC 2019 (成电杰出学生, **10/5000** in UESTC, **1/740** in department) 12/2019

iGEM "**Gold medal**" and "**Best Software Project**" (wiki: https://2019.igem.org/Team:UESTC-Software) 11/2019

Outstanding Graduates Award of UESTC 10/2019

"Wu Liang Ye" Enterprise Scholarship (**2/740** in Software Engineering department) 09/2019

RESEARCH EXPERIENCE

Research Assistant | Boston University, CIDAR lab Boston, MA | 09/2020-Present

Supervisor: Prof. Douglas Densmore

- Implement an online microfluidic device synthesis platform **"Neptune"** that formalizes a domain-specific language called "LFR", which can be compiled from a coding script to a visualizable design result as Verilog.
- Create prompt packages for 5 popular LLMs (Claude, GPT, Gemini, Qwen and DeepSeek) so that users can be assisted to use our synthesis automation platform without leaning a new programming language.
- Develop an in-silico validation algorithm **"Vespa"** for microfluidic experiment automation.
- Create and standardize a benchmark data set (synthesize + manual) for microfluidic device design algorithms.
- Formalize bio-related constraints and requirements to computational terms.
- Design and fabricate microfluidic devices supporting multiple experiment modes for a project aiming to research the strategy of cell-cell communication and conduct wet-lab experiments using these microfluidic devices.

Collaborative Research Assistant | MIT, Voigt lab Cambridge, MA | 11/2022-Present

Supervisor: Prof. Christopher Voigt

- Develop a new genetic circuit partitioning algorithm **"Oriole"** for the implementation of a cryptographic hash algorithm using a multi-cellular system, adhering to specific biological constraints.

iGEM Team Member | UESTC (Wiki: https://2019.igem.org/Team:UESTC-Software) Chengdu, China | 05-10/2019

- Developed an upgraded mapping method for DNA sequence to protein and demonstrated its superiority over other existing methods.
- Designed and implemented an innovative filtering model that significantly improved the accuracy of reflections between the iGEM registry and UniProt Database to over 95%. Additionally, the model successfully increased the number of these reflections by nearly 400%.

Student Research Internship | UCSB, Four "I"s lab Santa Barbara, CA | 06-08/2019

Supervisor: Prof. Matthew Turk and Prof. Tobies Hollerer

- Read through the paper talks about the existing best pose tracking algorithm and programmed it on local computer.
- Compared this algorithm with the new algorithm proposed by 2 PhD students using the same benchmark data set.

PUBLICATIONS

Google Scholar (https://scholar.google.com/citations?user=Zel-iSQAAAAJ&hl=en)

Journals:

[**J1**, IEEE-TCAD] **Zhou, Y.**, Oliveira, S. M., Sanka, R., McIntyre, D., & Densmore, D. (2024). Vespa: Logic-Level Constraint-Based Validation for Continuous-Flow Microfluidic Devices.
(Formal verification and constraint-based validation for large-scale microfluidic systems)

[**J2**, Nature Chemical Biology] Padmakumar, J. P., Sun, J. J., Cho, W., **Zhou, Y.**, Krenz, C., Han, W. Z., ... & Voigt, C. A. (2024). Partitioning of a 2-bit hash function across 66 communicating cells.
(Distributed multicellular biological computation and experimental validation)

[**J3**, IEEE-TCBB] **Zhou, Y.**, Voigt, C. A., & Densmore, D. (2025). Constraint-Based Sub-Graph Partitioning for Multi-Cellular Biological Networks.
(Graph-based optimization for scalable multicellular biological systems)

Conferences & Workshops:

[**C1**, IWBDA'22] A Conceptual Interactive Microfluidic Design and Control Workflow
[**C2**, IWBDA'25] LaNVis: Biological Constraint-based Large Network Visualizer (Accepted for oral presentation)

Manuscripts in Preparation:

[**J4**, in progress, targeting Nature Computational Science] *Neptune: Web-based Automated Design and Synthesis Platform for Microfluidic Devices.*
(LLM-enabled natural language design, formal grammar, automated synthesis for microfluidics)

PROFESSIONAL SERVICES & ACTIVITIES

International Workshop on Bio-Design Automation (IWBDA'25), **Oral Presentation**	09/2025
International Genetically Engineered Machine competition (iGEM'19, iGEM'24)	11/2019, 11/2024
Inventor/Contributor, **Patent: Agricultural Heavy Metal Biosensor-Integrated Device**	05-11/2024
Student Mentor of iGEM team: BostonU	04-10/2024
Chicago BioEngineering Conference (CBEC'24)	09/2024
Teaching Assistant, **EC504: Advanced Data Structures and Algorithms**, Boston University	01-05/2022, 01-05/2023
Engineering Biology Research Consortium (EBRC'23)	06/2023
International Workshop on Bio-Design Automation (IWBDA'22), Poster Presentation	10/2022

MENTORSHIP

Mentor, **EBRC Mentorship for Undergraduate and Master's Students Program (EMUMS)**	04/2025-Now
Direct Mentor, **Eric Xie**, Undergraduate Research Assistant, CIDAR Lab	06/2023-Now
Student Software Mentor, **iGEM Boston University Team** in 2024	04-10/2024

TECHNICAL SKILLS

AI & Data Science: PyTorch, TensorFlow, NetworkX
Systems & Software: Python, C/C++, Java, TypeScript, HTML, Linux Shell, LaTeX
Biological & Microfluidic Fabrication: CNC Milling, Laser Cutting
Design & Modeling Tools: Fusion 360, Microfluidic Devices Manual Design.